Manu Singh Burson

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SUMMARY

Computational social scientist with 7+ years of experience in: Causal Inference, Machine Learning, Bayesian Statistics, Forecasting, Survey Methods, Geo-spatial Data Analysis, and Experimental Methods; Substantive expertise in behavioral and media analytics.

EDUCATION

Columbia University Sep 2019 – Present
Ph.D., in Political Science (Comparative Politics and Statistical Methodology)
Columbia University May 2016
M.A, in Applied Statistics (Data Science Concentration)
University of Pune Jul 2012
B.E, in Mechanical Engineering (Distinction, Top 2.5% of graduating class)

EXPERIENCE

Research Fellow, The UN, Department of Peace Operations – New York Dec 2024 – Present

- Integrating data streams from 5+ social media monitoring APIs and digital news platforms into a comprehensive analysis pipeline for detecting and tracking disinformation campaigns targeting UN peacekeeping operations using Natural Language Processing (NLP) and open source LLM models.

Data Consultant, FiveThirtyEight, ABC News Network – New York Aug 2024 – Dec 2024

- Led comprehensive backtesting, sensitivity analysis, and code reviews of FiveThirtyEight's Bayesian election forecast model, identifying and resolving critical algorithmic vulnerabilities, resulting in numerical stability for the 2024 election models.

Graduate Researcher, Columbia University – New York Sep 2019 – Present

- Solo-author of novel analysis of political bot manipulation using Dynamic Difference-in-Differences, entity resolution, and NLP models on 382 Congressional Twitter accounts to reveal 30% bot-inflated visibility on traditional media; research won two prestigious APSA awards for methodological innovation.
- Analyzed 1TB of consumer behavior data using Regression Discontinuity in Time (RDiT), establishing a 17% increase in nationalistic goods consumption following right-wing sentiment surges, with findings presented at 3 international conferences.
- Co-developed a Bayesian Multidimensional Scaling model to estimate social group perceptions, revealing previously undetectable racial and religious bias.
- Conducted a large-scale evaluation of gender quotas in governance using fuzzy-matching algorithms to link 5,000+ local government units across multiple administrative datasets (election records, public works, census) spanning 2005-2023, producing novel evidence on the long-term impacts of women's representation.

Graduate Teaching Assistant, Columbia University – New York Sep 2019 – Present

- Conducted independent weekly discussion meetings to reinforce challenging concepts in statistics classes at the undergraduate, master's, and Ph.D. levels.
- Taught courses spanning Quantitative Methods 4, Applied Regression and Causal Inference, Bayesian Statistics, Advanced Topics in International Political Economy, and Principles of Quantitative Political Research.

Research Specialist II, Princeton University – Princeton Jun 2016 – Apr 2019

- Applied machine learning to create novel socioeconomic indicators for Afghanistan, combining satellite imagery, cellular network data, and topographic features, with results validated through field surveys.
- Created ensemble machine learning models for conflict prediction with 96% accuracy five years out used for informing early warning systems by the World Bank.
- Applied causal inference techniques and time-series analysis to quantify the impact of $26M in program investments across 398 districts in Afghanistan.

Predictive Analytics Fellow and Consultant, UN OCHA, Office for the Coordination of Humanitarian Affairs – The Hague and New York Jun 2018 – Aug 2022

- Selected from 700+ applicants for UN OCHA fellowship; developed a predictive model processing real-time data streams to forecast food insecurity and the socio-economic impact of COVID-19 in fragile contexts 6 months out; project secured a $500,000 grant from the Rockefeller Foundation.
- Implemented version control and peer review protocols for model deployment, reducing quality assurance cycle time.

Data Consultant, German Ministry of Economic Cooperation and Development – New York, NY — Feb 2018 – Jun 2020

- Primary data analyst responsible for evaluation of German development aid in Afghanistan using satellite imagery and multiple data sources (surveys, conflict data, aid records) to create metrics for development impact.

SKILLS

Programming: R (Advanced), Python, Stan

Research Design: Experimental Methods (A/B Testing, RCTs), Quasi-experimental Design, Survey Methodology

Statistical Analysis: Causal Inference (DiD, RDD, Matching), Bayesian Modeling, Time Series Forecasting

Advanced Analytics: Machine Learning, Natural Language Processing, High-dimensional Analytics, Spatial Statistics

Tools: ArcGIS/ArcPy, Data visualization, Git, SQL (basic), AWS/Hadoop (basic)

SELECTED PAPERS AND MEDIA (out of total 9)

- **Manu Singh, *Social media bots influence political news coverage***
  - Winner of the 2023 Best Student Paper Award in Information, Technology, and Politics at the American Political Science Association
  - Early Career Fellowship Award in the Elections, Public Opinion, and Voting Behavior section at the American Political Science Association
- Dejan Kovac, Manu Singh, Jacob N. Shapiro, *Public events boost nationalist identity* R&R at the Journal of Comparative Economics
- Donald Green, Manu Singh, Gaurav Sood, *New Evidence on the Effects of Randomly Assigned Reservations for Women Leaders in Indian Local Government*
- Andrew Gelman, Manu Singh, Yotam Margalit, David Halpbern, *Mapping mental identities using Bayesian multilevel modeling*

AWARDS AND GRANTS

Electoral Integrity Project: Junior Fellow and grant awardee	2023-2024
Society for Political Methodology Travel Award	2024
Experimental Laboratory for Social Sciences: Dissertation Grant, Columbia University	2020; 2023
Political Communication Section Travel Award	2023
SPARK Award Political Communication Section, APSA	2023
Dissertation Development Grant, Columbia University	2022; 2023
Dean's Fellowship, Columbia University	2019
Summer Institute in Computational Social Science, University of Edinburgh	2023
Weizenbaum Institute (Digital News Dynamics group), Visiting Fellowship	Expected Spring 2025

SELECTED INVITED PRESENTATIONS AND CONFERENCES

NYAAPOR & New York Open Statistical Programming Meetup	Oct 2024
Northeast Workshop in Empirical Political Science (NEWEPS 23)	Oct 2024
Meta's Computational Social Science Seminar	Oct 2024
American Political Science Association	Sep 2023; Sep 2024
Midwest Political Science Association	Apr 2022; Apr 2024
Society for Political Methodology	Jul 2024
ACM Conference on Economics and Computation, Yale University	Jun 2024
Empirical Studies of Conflict, Princeton University	Apr 2020, May 2021